

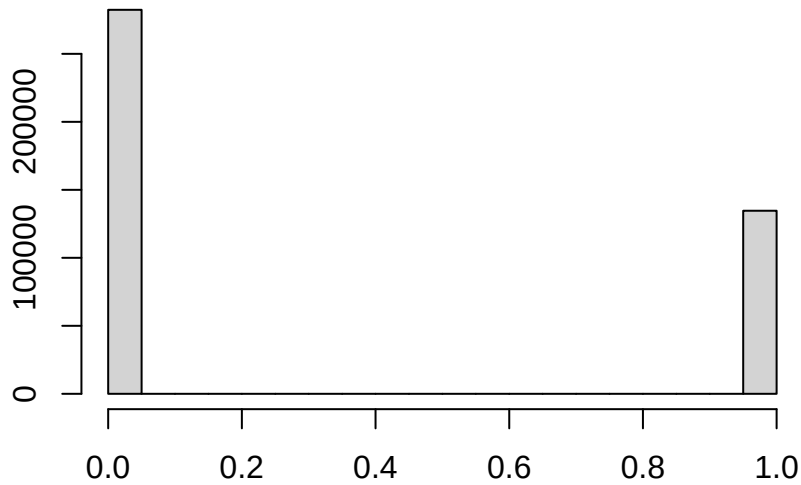
Competition – Contractor Experience

2024-06-12

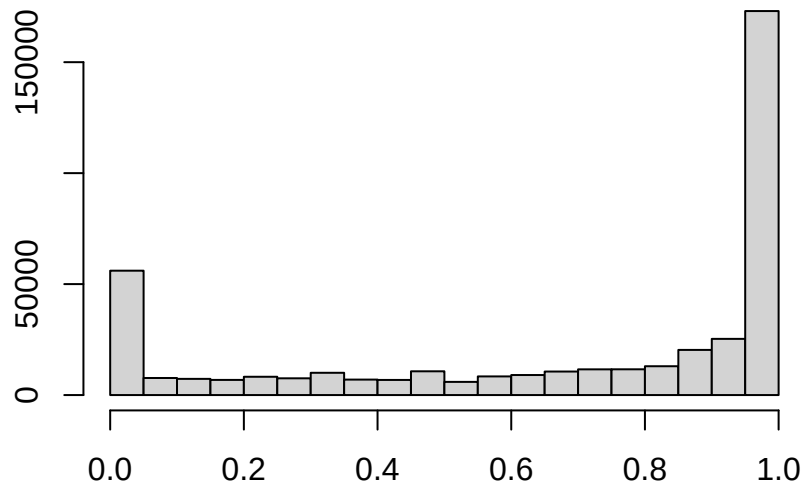
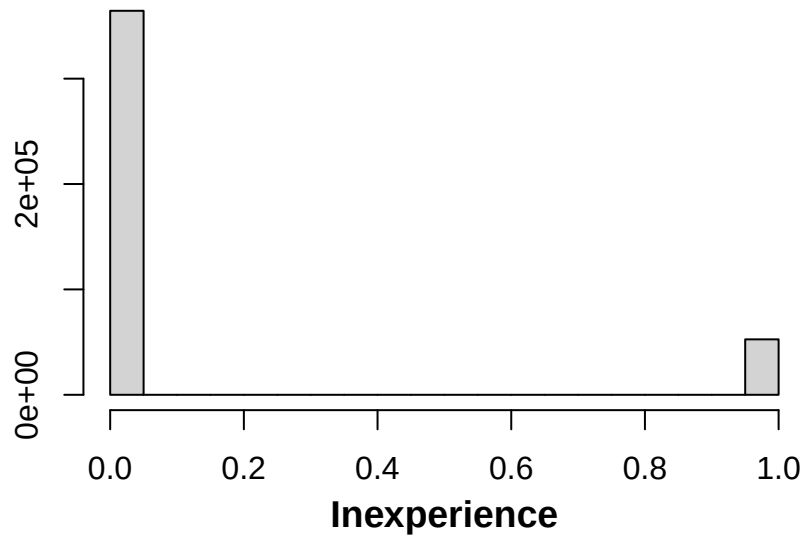
1 Time Invariant Status

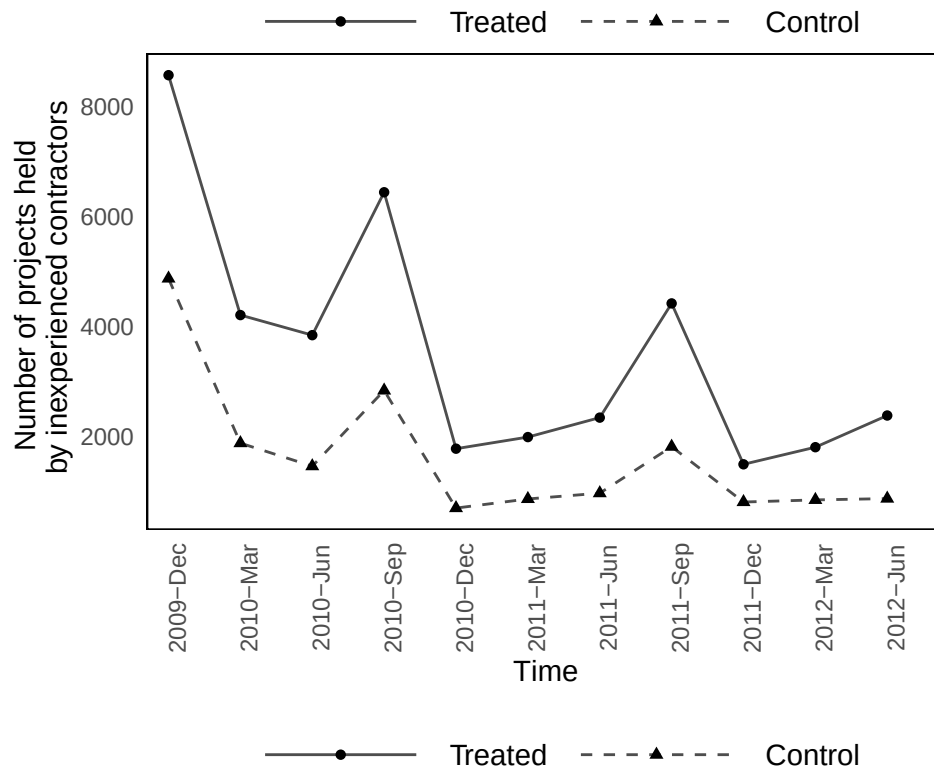
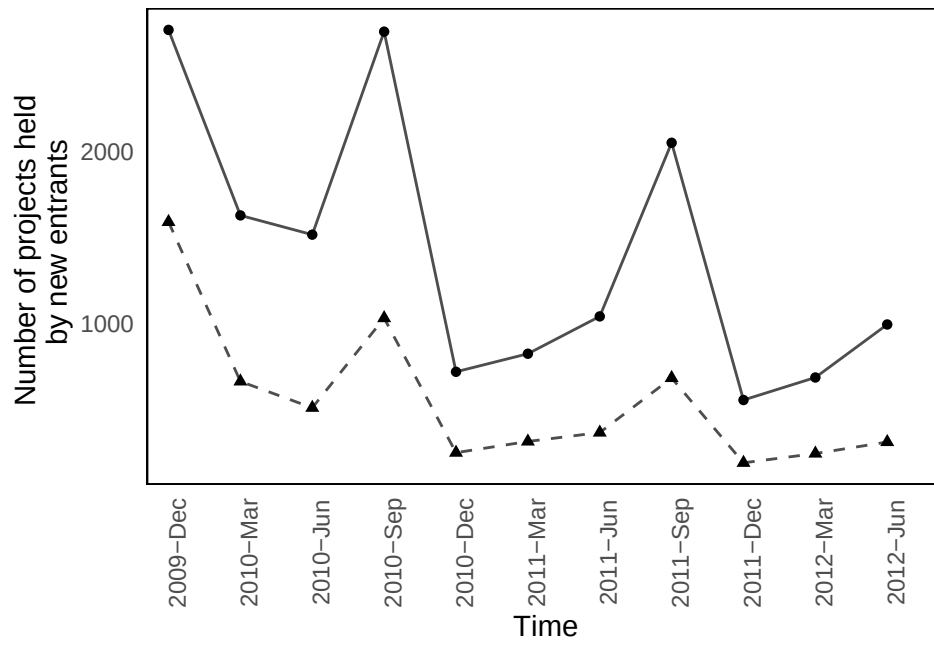
- Let n_i^{task} = Number of projects of the same task held by the contractor in the three quarters before starting project i
- Let n_i = Number of projects held by the contractor in the three quarters before starting project i
- Define:
 - Inexperienced contractor $_i$ = 1 if $n_i^{task} = 0$
 - New entrant $_i$ = 1 if $n_i = 0$
 - Level of inexperience $_i$ = $1 - \frac{n_i^{task}}{n_i}$ if $n_i > 0$
 - Level of inexperience $_i$ = 1 if $n_{i,k,t} = 0$.

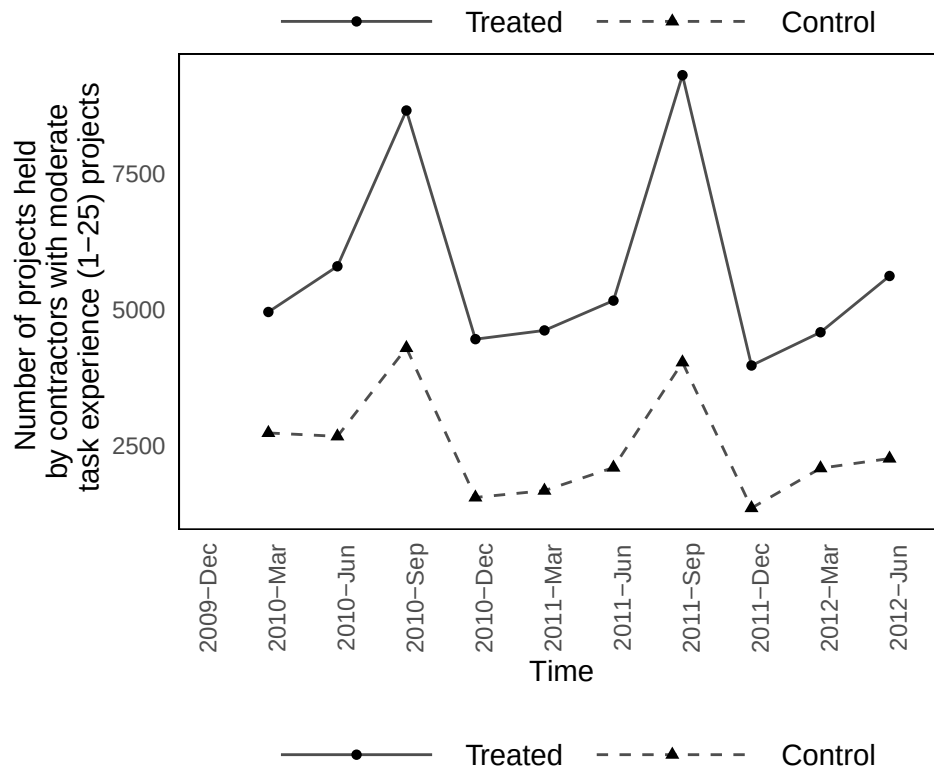
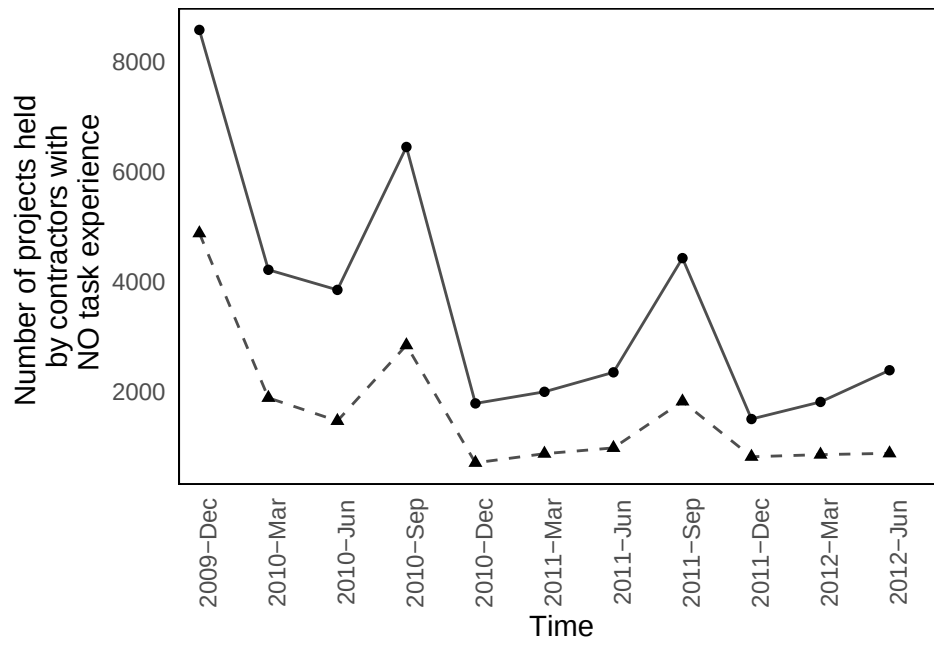
Task inexperience



New Entrant







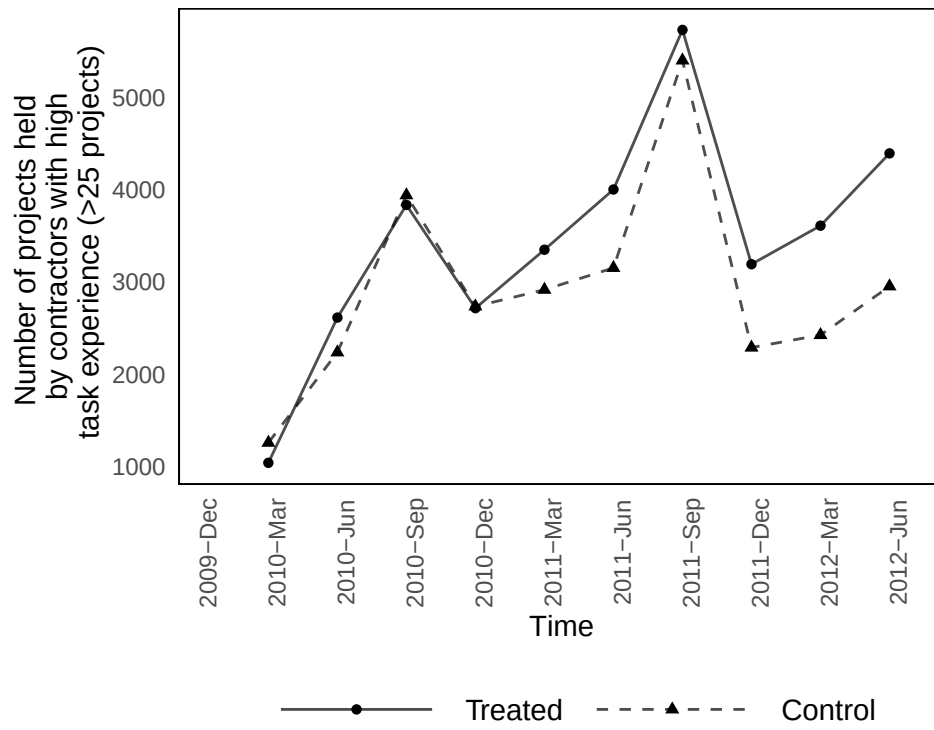


Table 1: Logit model

	Inexperienced contractor						
	I	II	III	IV	V	VI	VII
Constant	-0.495*** (0.013)	-0.039** (0.019)					
Treat	0.233*** (0.016)	0.273*** (0.022)	0.321*** (0.023)	0.493*** (0.029)	0.475*** (0.030)	0.403*** (0.031)	0.407*** (0.032)
Post	-0.759*** (0.015)	-0.842*** (0.019)					
Treat $\times$ Post	-0.010 (0.019)	0.007 (0.023)	-0.033 (0.023)	0.058** (0.028)	0.080*** (0.028)	0.090*** (0.031)	0.083*** (0.032)
Ln(Stage)		0.144*** (0.005)	0.188*** (0.005)	0.227*** (0.006)	0.229*** (0.006)	0.292*** (0.007)	0.297*** (0.007)
Num offers						0.019*** (0.003)	0.008** (0.004)
Ln(Initial Budget)						0.043*** (0.008)	0.038*** (0.008)
Ln(Initial Duration)						0.395*** (0.019)	0.538*** (0.022)
Post $\times$ Num offers							0.018*** (0.004)
Post $\times$ Ln(Initial Budget)							0.010 (0.007)
Post $\times$ Ln(Initial Duration)							-0.249*** (0.021)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	417,000	214,262	207,009	203,686	203,673	182,794	182,794
Squared Correlation	0.03	0.04	0.04	0.25	0.26	0.30	0.30
Pseudo R ²	0.03	0.03	0.03	0.21	0.22	0.25	0.25
BIC	510,459.90	271,116.63	256,631.10	211,169.48	209,754.79	177,324.96	177,101.39

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

2 Regression results

	Percentage delay rate						
	I	II	III	IV	V	VI	VII
Constant	6.938*** (0.152)	12.719*** (0.166)					
Treat	-3.861*** (0.176)	-3.484*** (0.166)	-3.486*** (0.166)	-1.834*** (0.158)	-1.873*** (0.157)	-1.161*** (0.138)	-1.007*** (0.139)
Inexperienced contractor	-0.329 (0.228)	-0.845*** (0.214)	-0.764*** (0.218)	-1.986*** (0.210)	-1.980*** (0.209)	0.776*** (0.176)	1.084*** (0.178)
Post	-0.401** (0.175)	-1.618*** (0.167)					
Treat $\times$ Inexperienced contractor	1.427*** (0.265)	1.681*** (0.250)	1.652*** (0.250)	0.798*** (0.235)	0.869*** (0.234)	-0.033 (0.204)	-0.151 (0.204)
Inexperienced contractor $\times$ Post	1.324*** (0.288)	1.112*** (0.274)	1.041*** (0.277)	0.009 (0.266)	-0.053 (0.265)	-0.017 (0.254)	-0.432* (0.256)
Treat $\times$ Post	1.325*** (0.207)	1.465*** (0.198)	1.481*** (0.198)	1.315*** (0.185)	1.339*** (0.183)	1.242*** (0.168)	1.014*** (0.173)
Treat $\times$ Inexperienced contractor $\times$ Post	-0.465 (0.343)	-0.991*** (0.328)	-0.977*** (0.328)	-0.428 (0.316)	-0.390 (0.315)	0.373 (0.312)	0.540* (0.310)
Ln(Stage)		3.657*** (0.034)	3.730*** (0.036)	3.562*** (0.034)	3.525*** (0.034)	3.182*** (0.032)	3.143*** (0.032)
Num offers						-0.056*** (0.009)	-0.111*** (0.013)
Ln(Initial Budget)						0.471*** (0.023)	0.535*** (0.031)
Ln(Initial Duration)						-9.134*** (0.096)	-10.113*** (0.126)
Post $\times$ Num offers							0.090*** (0.017)
Post $\times$ Ln(Initial Budget)							-0.099*** (0.034)
Post $\times$ Ln(Initial Duration)							1.593*** (0.162)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	214,195	214,148	214,148	214,148	214,148	192,629	192,629
R ²	0.01	0.07	0.07	0.14	0.14	0.27	0.27
Within R ²			0.07	0.06	0.06	0.19	0.19

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 2: Logit model

	I	II	III	New Entrant IV	V	VI	VII
Constant	-1.859*** (0.019)	-1.500*** (0.028)					
Treat	0.234*** (0.023)	0.230*** (0.031)	0.237*** (0.031)	0.579*** (0.039)	0.597*** (0.040)	0.557*** (0.042)	0.556*** (0.042)
Post	-0.546*** (0.023)	-0.567*** (0.027)					
Treat $\times$ Post	0.147*** (0.027)	0.178*** (0.032)	0.175*** (0.032)	0.180*** (0.036)	0.198*** (0.036)	0.224*** (0.040)	0.229*** (0.041)
Ln(Stage)		0.111*** (0.007)	0.131*** (0.007)	0.127*** (0.008)	0.125*** (0.008)	0.186*** (0.009)	0.186*** (0.009)
Num offers						0.022*** (0.004)	0.009* (0.005)
Ln(Initial Budget)						0.030*** (0.010)	0.020* (0.011)
Ln(Initial Duration)						0.291*** (0.024)	0.329*** (0.026)
Post $\times$ Num offers							0.024*** (0.005)
Post $\times$ Ln(Initial Budget)							0.018** (0.009)
Post $\times$ Ln(Initial Duration)							-0.075*** (0.028)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	417,000	214,262	214,262	210,777	210,748	189,340	189,340
Squared Correlation	0.01	0.01	0.02	0.17	0.19	0.20	0.20
Pseudo R ²	0.01	0.01	0.02	0.19	0.20	0.22	0.22
BIC	312,861.43	175,124.19	173,718.36	150,396.67	149,218.07	125,929.49	125,903.49

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

	Percentage delay rate						
	I	II	III	IV	V	VI	VII
Constant	6.946*** (0.125)	12.582*** (0.144)					
Treat	-3.573*** (0.145)	-3.168*** (0.136)	-3.172*** (0.136)	-1.772*** (0.133)	-1.772*** (0.133)	-1.141*** (0.116)	-1.006*** (0.118)
New Entrant	-0.958*** (0.295)	-1.369*** (0.277)	-1.324*** (0.278)	-1.592*** (0.276)	-1.610*** (0.276)	0.638*** (0.238)	0.873*** (0.240)
Post	-0.240 (0.148)	-1.426*** (0.142)					
Treat $\times$ New Entrant	2.277*** (0.350)	2.700*** (0.329)	2.657*** (0.329)	1.392*** (0.323)	1.427*** (0.323)	-0.111 (0.292)	-0.255 (0.293)
New Entrant $\times$ Post	1.843*** (0.395)	1.618*** (0.377)	1.571*** (0.377)	0.064 (0.373)	-0.024 (0.371)	0.245 (0.360)	-0.118 (0.362)
Treat $\times$ Post	1.192*** (0.175)	1.241*** (0.168)	1.256*** (0.168)	1.207*** (0.159)	1.222*** (0.158)	1.289*** (0.148)	1.085*** (0.152)
Treat $\times$ New Entrant $\times$ Post	-1.129** (0.470)	-1.788*** (0.452)	-1.754*** (0.451)	-0.694 (0.446)	-0.652 (0.445)	0.318 (0.451)	0.521 (0.449)
Ln(Stage)		3.666*** (0.034)	3.741*** (0.036)	3.516*** (0.034)	3.480*** (0.034)	3.204*** (0.032)	3.169*** (0.032)
Num offers						-0.055*** (0.009)	-0.109*** (0.013)
Ln(Initial Budget)						0.475*** (0.023)	0.542*** (0.031)
Ln(Initial Duration)						-9.106*** (0.096)	-10.056*** (0.124)
Post $\times$ Num offers							0.089*** (0.017)
Post $\times$ Ln(Initial Budget)							-0.104*** (0.034)
Post $\times$ Ln(Initial Duration)							1.551*** (0.161)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	214,195	214,148	214,148	214,148	214,148	192,629	192,629
R ²	0.01	0.07	0.07	0.14	0.14	0.27	0.27
Within R ²			0.07	0.06	0.06	0.19	0.19

Clustered (Project ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

3 Regressions – Continuous variable

	Level of inexperience						
	I	II	III	IV	V	VI	VII
Constant	0.726*** (0.002)	0.798*** (0.003)					
Treat	0.024*** (0.003)	0.035*** (0.003)	0.037*** (0.003)	0.011*** (0.003)	0.010*** (0.003)	-0.001 (0.003)	0.001 (0.003)
Post	-0.117*** (0.002)	-0.137*** (0.003)					
Treat × Post	0.007** (0.003)	0.005 (0.004)	0.004 (0.004)	0.020*** (0.003)	0.021*** (0.003)	0.019*** (0.003)	0.016*** (0.004)
Ln(Stage)		0.022*** (0.001)	0.025*** (0.001)	0.026*** (0.001)	0.026*** (0.001)	0.028*** (0.001)	0.028*** (0.001)
Num offers						0.004*** (0.000)	0.004*** (0.000)
Ln(Initial Budget)						0.001 (0.001)	0.003*** (0.001)
Ln(Initial Duration)						0.027*** (0.002)	0.037*** (0.002)
Post × Num offers							0.001 (0.000)
Post × Ln(Initial Budget)							-0.003*** (0.001)
Post × Ln(Initial Duration)							-0.016*** (0.002)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	417,000	214,262	214,262	214,262	214,262	192,743	192,743
R ²	0.02	0.04	0.06	0.28	0.29	0.31	0.31
Within R ²			0.01	0.01	0.01	0.02	0.02

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

	Percentage delay rate						
	I	II	III	IV	V	VI	VII
Constant	5.081*** (0.250)	11.347*** (0.249)					
Treat	-0.844*** (0.317)	-0.761** (0.300)	-0.811*** (0.300)	0.375 (0.282)	0.248 (0.280)	-0.221 (0.262)	-0.188 (0.263)
Level of inexperience	2.242*** (0.315)	1.334*** (0.296)	1.409*** (0.300)	0.454 (0.283)	0.463 (0.282)	1.938*** (0.253)	2.172*** (0.253)
Post	-0.200 (0.286)	-1.471*** (0.273)					
Treat $\times$ Level of inexperience	-3.010*** (0.386)	-2.485*** (0.364)	-2.439*** (0.364)	-2.473*** (0.339)	-2.312*** (0.337)	-1.158*** (0.309)	-1.063*** (0.309)
Level of inexperience $\times$ Post	0.778** (0.373)	0.727** (0.355)	0.656* (0.358)	-0.175 (0.340)	-0.156 (0.340)	-0.541* (0.322)	-0.771** (0.321)
Treat $\times$ Post	0.765** (0.367)	1.099*** (0.351)	1.152*** (0.351)	0.531 (0.329)	0.651** (0.327)	0.659** (0.318)	0.667** (0.319)
Treat $\times$ Level of inexperience $\times$ Post	-0.288 (0.463)	-0.795* (0.443)	-0.831* (0.443)	0.279 (0.420)	0.150 (0.418)	0.805** (0.405)	0.553 (0.403)
Ln(Stage)		3.667*** (0.034)	3.741*** (0.035)	3.523*** (0.034)	3.486*** (0.034)	3.181*** (0.032)	3.143*** (0.032)
Num offers						-0.057*** (0.009)	-0.114*** (0.013)
Ln(Initial Budget)						0.475*** (0.023)	0.536*** (0.030)
Ln(Initial Duration)						-9.111*** (0.096)	-10.080*** (0.125)
Post $\times$ Num offers							0.092*** (0.017)
Post $\times$ Ln(Initial Budget)							-0.095*** (0.034)
Post $\times$ Ln(Initial Duration)							1.581*** (0.161)
Time FE	No	No	Yes	Yes	Yes	Yes	Yes
Task FE	No	No	No	Yes	Yes	Yes	Yes
Industry FE	No	No	No	No	Yes	Yes	Yes
Observations	214,195	214,148	214,148	214,148	214,148	192,629	192,629
R ²	0.01	0.07	0.07	0.14	0.14	0.27	0.27
Within R ²			0.07	0.06	0.06	0.19	0.19

Clustered (Project ID) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1